

3.6.2 Cyber security threats

Content	Additional information
Understand and be able to explain the following cyber security threats: • social engineering techniques • malicious code (malware) • pharming • weak and default passwords • misconfigured access rights • removable media • unpatched and/or outdated software.	Pharming is a cyber attack intended to redirect a website's traffic to a fake website.
Explain what penetration testing is and what it is used for.	Penetration testing is the process of attempting to gain access to resources without knowledge of usernames, passwords and other normal means of access. Students should understand the following two types of penetration testing: • when the person or team testing the system has knowledge of and possibly basic credentials for the target system, simulating an attack from inside the system (a malicious insider) • when the person or team testing the system has no knowledge of any credentials for the target system, simulating an attack from outside the system (an external attack).

3.6.2.1 Social engineering

Content	Additional information
Define the term social engineering. Describe what social engineering is and how it can be protected against. Explain the following forms of social engineering: • blagging (pretexting) • phishing • shouldering (or shoulder surfing).	Students should know that social engineering is the art of manipulating people so they give up confidential information. Blagging is the act of creating and using an invented scenario to engage a targeted victim in a manner that increases the chance the victim will divulge information or perform actions that would be unlikely in ordinary circumstances. Phishing is a technique of fraudulently obtaining private information, often using email or SMS. Shouldering is observing a person's private information over their shoulder eg cashpoint machine PIN numbers.

3.6.2.2 Malicious code (malware)

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Define the term malware. Describe what malware is and how it can be protected against. Describe the following forms of malware: • computer virus • trojan • spyware.	Malware is an umbrella term used to refer to a variety of forms of hostile or intrusive software.